

Ghosh's Inequality

A Quartic Inequality in Three Nonnegative Variables

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Abstract

We study the symmetric homogeneous quartic inequality

$$a^4 + b^4 + c^4 \geq abc(a + b + c),$$

for nonnegative real numbers a, b, c . A direct proof follows from the weighted arithmetic–geometric mean inequality. We characterize the equality case, establish the optimality of the coefficient on the right-hand side, derive normalized and ratio formulations, and give several immediate corollaries. An interpretation in exponent space shows that the proof is naturally associated with convex combinations of the exponent vectors of the pure quartic monomials.

The paper ends with “The End”

1 Introduction

Let a, b, c be nonnegative real numbers. We consider the inequality

$$a^4 + b^4 + c^4 \geq abc(a + b + c).$$

Expanding the right-hand side gives

$$abc(a + b + c) = a^2bc + ab^2c + abc^2.$$

Thus the inequality is equivalently

$$a^4 + b^4 + c^4 \geq a^2bc + ab^2c + abc^2.$$

The inequality has two elementary but important structural properties. First, it is symmetric in a, b, c . Second, both sides are homogeneous of degree four. Consequently, for every $\lambda \geq 0$, replacing (a, b, c) by $(\lambda a, \lambda b, \lambda c)$ multiplies both sides by λ^4 .

The principal result is the following.

Theorem 1 (Ghosh's Inequality). *For all nonnegative real numbers a, b, c ,*

$$a^4 + b^4 + c^4 \geq abc(a + b + c).$$

Equality holds if and only if

$$a = b = c.$$

The proof is an exact application of weighted AM–GM. Each mixed quartic monomial on the right-hand side is bounded by a suitable weighted average of a^4, b^4, c^4 , and the resulting coefficients add to exactly one for each pure quartic term.

2 Preliminary AM–GM Lemma

We recall the weighted arithmetic–geometric mean inequality.

Lemma 1 (Weighted AM–GM). *Let $x_1, \dots, x_n$ be positive real numbers, and let $\lambda_1, \dots, \lambda_n$ be nonnegative real numbers satisfying*

$$\lambda_1 + \dots + \lambda_n = 1.$$

Then

$$\lambda_1 x_1 + \dots + \lambda_n x_n \geq x_1^{\lambda_1} \dots x_n^{\lambda_n}.$$

If all λ_i are positive, equality holds if and only if

$$x_1 = x_2 = \dots = x_n.$$

The result extends to nonnegative variables by continuity whenever all expressions involved are well defined.

3 Proof of Ghosh’s Inequality

Proof. Weighted AM–GM applied to a^4, b^4, c^4 with weights

$$\frac{1}{2}, \quad \frac{1}{4}, \quad \frac{1}{4}$$

gives

$$\begin{aligned} \frac{1}{2}a^4 + \frac{1}{4}b^4 + \frac{1}{4}c^4 &\geq (a^4)^{1/2}(b^4)^{1/4}(c^4)^{1/4} \\ &= a^2bc. \end{aligned}$$

Therefore,

$$a^2bc \leq \frac{1}{2}a^4 + \frac{1}{4}b^4 + \frac{1}{4}c^4. \tag{1}$$

Cyclically,

$$ab^2c \leq \frac{1}{4}a^4 + \frac{1}{2}b^4 + \frac{1}{4}c^4, \tag{2}$$

and

$$abc^2 \leq \frac{1}{4}a^4 + \frac{1}{4}b^4 + \frac{1}{2}c^4. \tag{3}$$

Adding (1), (2), and (3), we obtain

$$\begin{aligned} a^2bc + ab^2c + abc^2 &\leq \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{4}\right)a^4 \\ &\quad + \left(\frac{1}{4} + \frac{1}{2} + \frac{1}{4}\right)b^4 \\ &\quad + \left(\frac{1}{4} + \frac{1}{4} + \frac{1}{2}\right)c^4 \\ &= a^4 + b^4 + c^4. \end{aligned}$$

Since

$$a^2bc + ab^2c + abc^2 = abc(a + b + c),$$

we conclude that

$$\boxed{a^4 + b^4 + c^4 \geq abc(a + b + c).}$$

□

4 Equivalent Four-Term AM–GM Proof

The same argument may be expressed using only the ordinary, unweighted AM–GM inequality.

For the four nonnegative quantities

$$a^4, \quad a^4, \quad b^4, \quad c^4,$$

AM–GM gives

$$\frac{2a^4 + b^4 + c^4}{4} \geq \sqrt[4]{a^8 b^4 c^4}.$$

Since $a, b, c \geq 0$,

$$\sqrt[4]{a^8 b^4 c^4} = a^2 bc.$$

Hence

$$2a^4 + b^4 + c^4 \geq 4a^2 bc. \quad (4)$$

Cyclically,

$$a^4 + 2b^4 + c^4 \geq 4ab^2 c, \quad (5)$$

and

$$a^4 + b^4 + 2c^4 \geq 4abc^2. \quad (6)$$

Adding (4), (5), and (6) yields

$$4a^4 + 4b^4 + 4c^4 \geq 4a^2 bc + 4ab^2 c + 4abc^2.$$

Dividing by 4 gives

$$a^4 + b^4 + c^4 \geq a^2 bc + ab^2 c + abc^2 = abc(a + b + c).$$

Thus the weighted and unweighted AM–GM proofs are the same argument written in two equivalent forms.

5 Equality Conditions

We now determine the equality case carefully.

Suppose first that

$$a = b = c = t$$

for some $t \geq 0$. Then

$$a^4 + b^4 + c^4 = 3t^4,$$

while

$$abc(a + b + c) = t^3(3t) = 3t^4.$$

Thus equality holds whenever $a = b = c$.

Conversely, suppose equality holds in Ghosh's inequality. Since the proof was obtained by adding the three nonnegative AM–GM gaps in (1)–(3), equality in the final inequality requires equality in each of those three inequalities.

If $a, b, c > 0$, equality in (1) requires

$$a^4 = b^4 = c^4.$$

Because a, b, c are nonnegative, this implies

$$a = b = c.$$

It remains to consider the boundary. If at least one of a, b, c is zero, then

$$abc(a + b + c) = 0.$$

Equality would therefore require

$$a^4 + b^4 + c^4 = 0.$$

Since every term is nonnegative, this is possible only when

$$a = b = c = 0.$$

Consequently, over the entire nonnegative orthant,

$$\boxed{a^4 + b^4 + c^4 = abc(a + b + c) \iff a = b = c.}$$

6 Homogeneity and Normalization

Define the quartic form

$$F(a, b, c) = a^4 + b^4 + c^4 - abc(a + b + c).$$

For every $\lambda \geq 0$,

$$\begin{aligned} F(\lambda a, \lambda b, \lambda c) &= \lambda^4 a^4 + \lambda^4 b^4 + \lambda^4 c^4 - \lambda^4 abc(a + b + c) \\ &= \lambda^4 F(a, b, c). \end{aligned}$$

Hence F is homogeneous of degree four.

This permits several convenient normalizations.

6.1 Normalization by the sum

If $a + b + c > 0$, set

$$x = \frac{a}{a + b + c}, \quad y = \frac{b}{a + b + c}, \quad z = \frac{c}{a + b + c}.$$

Then

$$x + y + z = 1.$$

By homogeneity, Ghosh's inequality is equivalent to

$$x^4 + y^4 + z^4 \geq xyz.$$

Thus, under the normalization

$$a + b + c = 1,$$

the inequality becomes simply

$$\boxed{a^4 + b^4 + c^4 \geq abc.}$$

6.2 Normalization by the product

If $a, b, c > 0$, homogeneity also permits the normalization

$$abc = 1.$$

Under this condition,

$$abc(a + b + c) = a + b + c,$$

and therefore

$$\boxed{a^4 + b^4 + c^4 \geq a + b + c.}$$

7 Scale-Invariant Ratio

For $a, b, c > 0$, define

$$R(a, b, c) = \frac{a^4 + b^4 + c^4}{abc(a + b + c)}.$$

The theorem states exactly that

$$R(a, b, c) \geq 1.$$

Moreover, for every $\lambda > 0$,

$$\begin{aligned} R(\lambda a, \lambda b, \lambda c) &= \frac{\lambda^4(a^4 + b^4 + c^4)}{\lambda^4 abc(a + b + c)} \\ &= R(a, b, c). \end{aligned}$$

Thus R is homogeneous of degree zero and is therefore invariant under changes of scale.

The equality characterization implies

$$\boxed{\min_{a, b, c > 0} R(a, b, c) = 1,}$$

with the minimum attained precisely when

$$a = b = c > 0.$$

8 Optimality of the Constant

A natural generalization is

$$a^4 + b^4 + c^4 \geq Kabc(a + b + c),$$

where K is a real constant independent of a, b, c .

Proposition 1. *The largest real constant K such that*

$$a^4 + b^4 + c^4 \geq Kabc(a + b + c)$$

for all $a, b, c \geq 0$ is

$$\boxed{K = 1}.$$

Proof. Ghosh's inequality establishes that $K = 1$ is admissible.

To show that no larger constant is possible, set

$$a = b = c = t,$$

where $t > 0$. The proposed inequality becomes

$$3t^4 \geq 3Kt^4.$$

Since $3t^4 > 0$, division gives

$$1 \geq K.$$

Hence every admissible constant satisfies $K \leq 1$.

Since $K = 1$ is admissible, it is the largest possible constant. □

Thus the coefficient on the right-hand side of Ghosh's inequality is sharp.

9 Exponent-Space Interpretation

The weighted AM–GM proof has a natural geometric interpretation.

Associate with a monomial

$$a^\alpha b^\beta c^\gamma$$

its exponent vector

$$(\alpha, \beta, \gamma).$$

The pure quartic monomials correspond to

$$a^4 \longleftrightarrow (4, 0, 0),$$

$$b^4 \longleftrightarrow (0, 4, 0),$$

and

$$c^4 \longleftrightarrow (0, 0, 4).$$

The mixed monomial a^2bc corresponds to

$$(2, 1, 1).$$

But

$$(2, 1, 1) = \frac{1}{2}(4, 0, 0) + \frac{1}{4}(0, 4, 0) + \frac{1}{4}(0, 0, 4).$$

The coefficients form a convex combination because

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1.$$

Weighted AM–GM converts this convex relation among exponent vectors directly into

$$a^2bc \leq \frac{1}{2}a^4 + \frac{1}{4}b^4 + \frac{1}{4}c^4.$$

Similarly,

$$(1, 2, 1) = \frac{1}{4}(4, 0, 0) + \frac{1}{2}(0, 4, 0) + \frac{1}{4}(0, 0, 4),$$

which gives

$$ab^2c \leq \frac{1}{4}a^4 + \frac{1}{2}b^4 + \frac{1}{4}c^4,$$

and

$$(1, 1, 2) = \frac{1}{4}(4, 0, 0) + \frac{1}{4}(0, 4, 0) + \frac{1}{2}(0, 0, 4),$$

which gives

$$abc^2 \leq \frac{1}{4}a^4 + \frac{1}{4}b^4 + \frac{1}{2}c^4.$$

The three mixed exponent vectors therefore lie in the convex hull of the three pure quartic exponent vectors. The coefficient balance is exactly what permits the three AM–GM bounds to sum to the original inequality.

10 Matrix Form of the Coefficient Allocation

The coefficient allocation in the weighted AM–GM proof may be summarized by the matrix

$$W = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{pmatrix}.$$

The first row controls a^2bc , the second controls ab^2c , and the third controls abc^2 .

Every row sum equals one:

$$W \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Likewise, every column sum equals one. Thus W is doubly stochastic.

Indeed,

$$W = \frac{1}{4} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

The column sums explain why, after the three AM–GM inequalities are added, the total coefficient attached to each of a^4, b^4, c^4 is exactly one.

This gives an elementary linear-algebraic representation of the coefficient bookkeeping underlying the proof.

11 Immediate Corollaries

Corollary 1. For $a, b, c > 0$,

$$\frac{a^3}{bc} + \frac{b^3}{ca} + \frac{c^3}{ab} \geq a + b + c.$$

Equality holds if and only if $a = b = c$.

Proof. Since $abc > 0$, divide Ghosh's inequality by abc :

$$\frac{a^4 + b^4 + c^4}{abc} \geq a + b + c.$$

Expanding the left-hand side gives

$$\frac{a^3}{bc} + \frac{b^3}{ca} + \frac{c^3}{ab} \geq a + b + c.$$

The equality condition follows directly from the theorem. □

Corollary 2. *If $a, b, c \geq 0$ and*

$$a + b + c = 1,$$

then

$$a^4 + b^4 + c^4 \geq abc.$$

Equality holds if and only if

$$a = b = c = \frac{1}{3}.$$

Proof. Since $a + b + c = 1$, Ghosh's inequality gives

$$a^4 + b^4 + c^4 \geq abc(a + b + c) = abc.$$

Equality in the original inequality requires $a = b = c$, and the normalization then gives $a = b = c = 1/3$. $\square$

Corollary 3. *If $a, b, c > 0$ and*

$$abc = 1,$$

then

$$a^4 + b^4 + c^4 \geq a + b + c.$$

Equality holds if and only if

$$a = b = c = 1.$$

Proof. The condition $abc = 1$ gives

$$abc(a + b + c) = a + b + c.$$

Hence Ghosh's inequality yields

$$a^4 + b^4 + c^4 \geq a + b + c.$$

Equality requires $a = b = c$, and $abc = 1$ then implies $a = b = c = 1$. $\square$

12 A General Monomial-Domination Principle

The proof is an instance of a useful general principle. Let $\alpha, \beta, \gamma \geq 0$ satisfy

$$\alpha + \beta + \gamma = 4.$$

Then

$$a^\alpha b^\beta c^\gamma = (a^4)^{\alpha/4} (b^4)^{\beta/4} (c^4)^{\gamma/4}.$$

Weighted AM–GM therefore gives

$$a^\alpha b^\beta c^\gamma \leq \frac{\alpha}{4} a^4 + \frac{\beta}{4} b^4 + \frac{\gamma}{4} c^4.$$

For $(\alpha, \beta, \gamma) = (2, 1, 1)$, this becomes

$$a^2 bc \leq \frac{1}{2} a^4 + \frac{1}{4} b^4 + \frac{1}{4} c^4.$$

The cyclic permutations $(1, 2, 1)$ and $(1, 1, 2)$ yield the other two terms required for Ghosh's inequality.

Thus the theorem can be regarded as the sum of three instances of a general domination rule for homogeneous monomials of degree four.

13 Numerical Illustrations

The following examples illustrate the theorem.

a	b	c	$a^4 + b^4 + c^4$	$abc(a + b + c)$
1	1	1	3	3
2	1	1	18	8
1	2	3	98	36
2	2	2	48	48
1	1	0	2	0
0	0	0	0	0

The equal-variable triples attain equality. Nonzero asymmetric triples in the table yield strict inequality, while boundary points with one zero variable yield strict inequality unless all three variables are zero.

14 Discussion

The inequality

$$a^4 + b^4 + c^4 \geq abc(a + b + c)$$

has a particularly balanced structure. Its left-hand side consists of the three pure quartic monomials, while its right-hand side consists of the three symmetric mixed quartic monomials

$$a^2bc, \quad ab^2c, \quad abc^2.$$

The weighted AM–GM proof shows that the coefficient balance is exact. For each mixed term, one variable receives weight $1/2$ and the other two receive weight $1/4$. Summing cyclically causes every pure fourth power to receive total weight

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1.$$

This also explains the sharpness of the inequality. No coefficient is lost in the summation, and the symmetric point $a = b = c$ simultaneously saturates all three AM–GM inequalities.

The inequality therefore provides a simple example of a broader phenomenon: homogeneous mixed monomials may be controlled by pure powers when their exponent vectors admit suitable convex decompositions and the corresponding weights can be balanced across the desired sum.

15 Conclusion

For all nonnegative real numbers a, b, c ,

$$a^4 + b^4 + c^4 \geq abc(a + b + c).$$

The result follows immediately from the three weighted AM–GM bounds

$$a^2bc \leq \frac{1}{2}a^4 + \frac{1}{4}b^4 + \frac{1}{4}c^4,$$

$$ab^2c \leq \frac{1}{4}a^4 + \frac{1}{2}b^4 + \frac{1}{4}c^4,$$

and

$$abc^2 \leq \frac{1}{4}a^4 + \frac{1}{4}b^4 + \frac{1}{2}c^4.$$

Adding them gives exactly

$$abc(a + b + c) \leq a^4 + b^4 + c^4.$$

Equality holds precisely when

$$a = b = c,$$

including the boundary equality point $(0, 0, 0)$. Furthermore, the constant 1 multiplying $abc(a + b + c)$ is optimal.

The proof may be understood analytically through AM–GM, geometrically through convex combinations of exponent vectors, and algebraically through a doubly stochastic coefficient-allocation matrix. These equivalent viewpoints expose the simple structural reason underlying the inequality.

References

- [1] G. H. Hardy, J. E. Littlewood, and G. Pólya, *Inequalities*, Cambridge University Press, Cambridge, 1934.
- [2] D. S. Mitrinović, *Analytic Inequalities*, Springer-Verlag, Berlin, 1970.
- [3] J. M. Steele, *The Cauchy–Schwarz Master Class: An Introduction to the Art of Mathematical Inequalities*, Cambridge University Press, Cambridge, 2004.

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